See talk EGU26-19117 -Sunspot Number V2.0 Through Solar Cycle 25: A Long-term Multi-Proxy Stability Analysis.
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Reconstruction of the Sunspot Number Series : Gathering Data

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Abstract

The Sunspot Number (S_N) is one of the longest continuous records of solar activity. S_N V2.0 significantly improved the historical series through targeted recalibrations, but recent studies have shown that some scale changes—such as the 1849 transition—can only be fully understood by returning to the original raw observations. This motivates a complete reconstruction (S_N V3.0) in which uncertainties and calibration steps arise directly from the quality of each dataset rather than from global statistical corrections.

To enable this reconstruction, the historical dataset has expanded dramatically in recent years. WDC–SILSO digitized the Mittheilungen (1610–1944) between 2017 and 2019, converting Wolf's original Zürich journals into machine-readable datasets with modern metadata and provenance. As part of FARSUN, the Zürich observation tables for 1945–1979 were digitized from 2023 to 2025, completing the recovery of the full Zürich legacy. In parallel, major community efforts (e.g., Hayakawa, Arlt, Vaquero, Carrasco, Vokhmyanin Zolotova) have located, transcribed, and published numerous additional early telescopic sunspot records since 2010.

Together, these datasets greatly expand the number of overlapping observations and allow independent cross-checks across observers and epochs. This richer observational base strengthens scale-transfer analyses, improves calibration robustness, and provides the foundation for an uncertainty-quantified reconstruction of the long-term sunspot series. These advances make a defensible S_N V3.0 possible and ensure that the Sunspot Number remains a reliable reference for solar activity studies.

Methodology

1. **Collect Sources**: gather historical observing records from many archives (e.g. Wolf's *Mittheilungen*, Zürich tables, manuscripts, drawings) and record where each piece of data comes from.
2. **Digitize Data**: convert handwritten and printed records into digital form using handwriting recognition tools and careful manual entry when needed.
3. **Check & Align**: verify the data, resolve different names for the same observers, align dates and formats, and account for how different observers counted sunspots.
4. **Share Results**: provide the data in easy-to-use formats, including online access tools (APIs), and include uncertainty estimates so users can assess how reliable the data are.

FARSuN Historical Sunspot Database

A sustainable, interoperable reference database ensuring that four centuries of solar observations remain accessible and scientifically reliable for the next generation of Sun-climate research.

- Legacy sources (Mitteilungen, Wolf Institute, etc.) have been integrated into the current schema for cross-validation and quality control.
- Automated quality checks detect inconsistencies (e.g., NG > NS, duplicates, missing metadata).
- Metadata harmonization for observers, instruments, and observatories is underway.

Next steps include finalizing the data schema and completing links to legacy data sets, expanding automated quality-control (QC) and improving uncertainty modeling, and increasing digitization and data extraction (Adams, Gruithuisen, Stark, etc.) using handwriting recognition (HTR) and citizen-science contributions. The full data set will be made available through the EPN-TAP service, following FAIR (Findable, Accessible, Interoperable, Reusable) principles. This harmonized database will then support the S_N V3.0 calibration and long-term solar activity studies.

Digitize and Ingest data

- Transcription: combine handwriting recognition (OCR) with manual entry for complex tables.
- OCR on existing online Mittheilungen journals to extract metadata automatically.
- Combine extracted data into homogeneous format to ingest into the database.

Quality Control

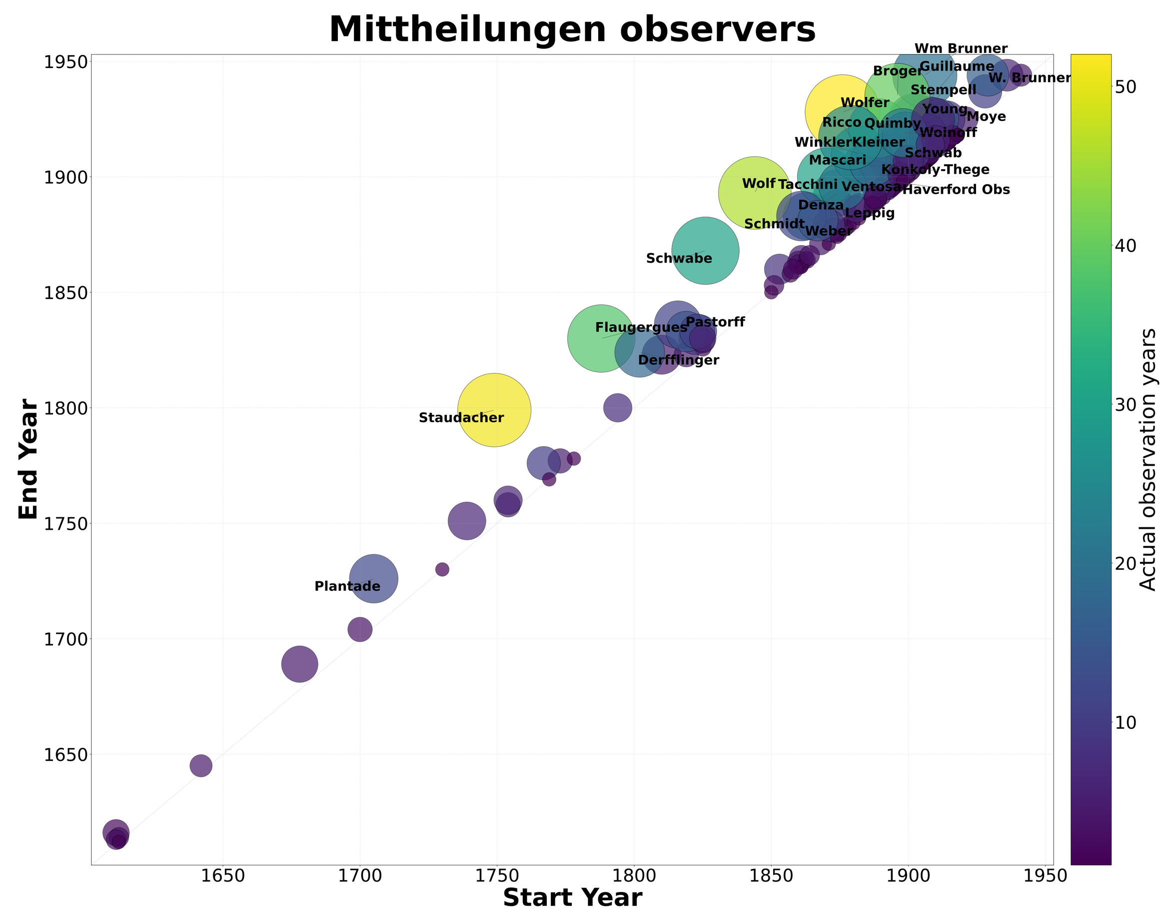
- Verification across same time observers: e.g. counts from Adams (1819-1823) and Gruithuisen (1817-1849) are checked independently before deriving indices.
- Cross-checks: compare FARSuN transcriptions with Wolf's *Mittheilungen* and modern SILSO indices to detect inconsistencies over time.
- Metadata consistency: ensure each observer has a unique ID, and preserve information on instruments, location, and observing conditions.
- Reproducibility: track all data processing steps with Git and notebooks so results can be checked and reused.

Key Datasets

Mittheilungen Dataset (1610–1944)

The Mittheilungen journals, compiled by Rudolf Wolf and successors at the Zürich Observatory from 1849 onwards, form a central source of historical sunspot observations. They document daily group and spot counts from 1848 to 1944 and incorporate earlier material back to 1610 through Wolf's own research into archives. Within the FARSuN project, the complete series was digitized and structured at the Royal Observatory of Belgium, integrated into the historical sunspot database, and is undergoing continued quality checks, metadata harmonization, and cross-comparisons with newly recovered records. This digitized dataset now provides four centuries of observer statistics and constitutes a key component of the historical foundation for the ongoing reconstruction of S_N V3.0.

- Refining data consistency and observer calibration (e.g. Wolf–Wolfer overlap).
- Finalizing metadata alignment for FAIR publication.
- Resolving remaining ID and provenance issues across merged databases.

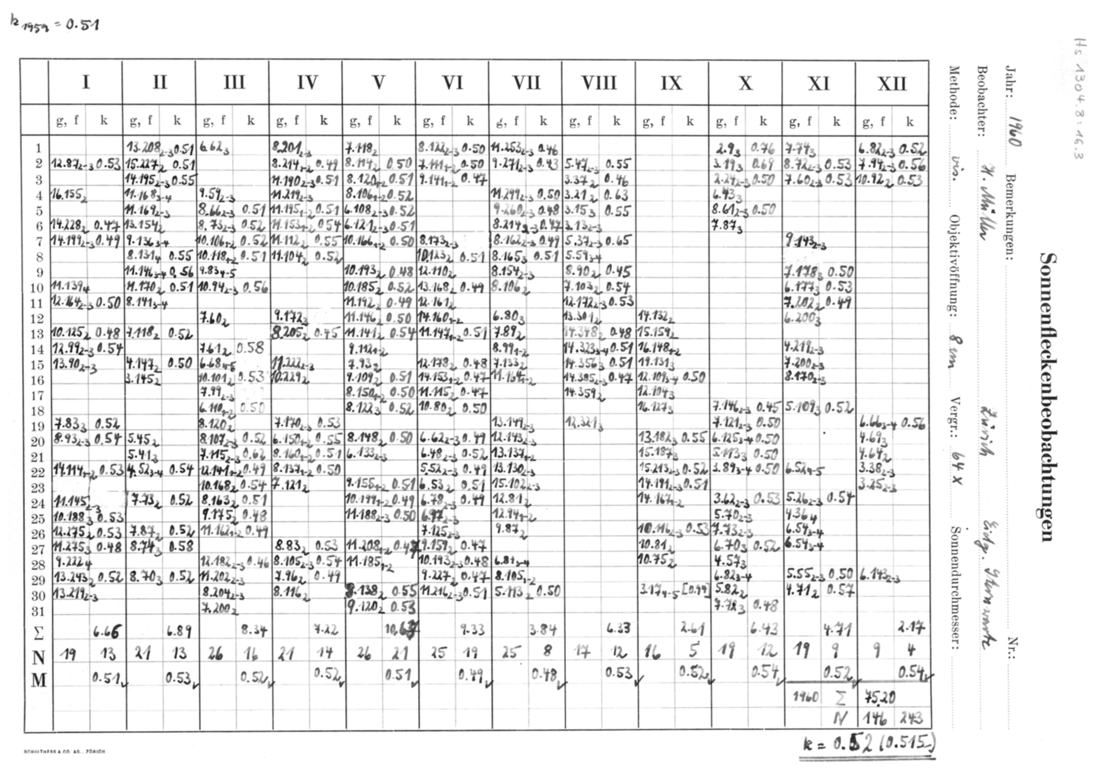


Zürich Tables (1945–1979)

The Zürich Tables form the foundational record of daily sunspot number production at the Zürich Observatory between 1945 and 1979, detailing group and spot counts, the adopted Wolf number, and observing context.

Status: All 1500+ yearly tables are digitized and the remaining observer metadata are being encoded.

- Fully digitized for 1945–1979 with detailed metadata completed for 1945–1959.
- Quality control and observer metadata harmonisation underway to maintain traceability.



A typical handwritten yearly table from the Zürich archives.

Gruithuisen Data (1817–1849)

The Gruithuisen manuscripts (1817–1849) capture daily sunspot notes, drawings, and weather context in German Kurrent script, supplying early-19th-century coverage vital for S_N V3.0 refinement. The entire image set is digitized and extraction via different OCR models and quality control for NG/NS counts is underway.

- Image sets and text snippets fully extracted and collected with preliminary CSV/XLSX extraction completed.
- Two-pass quality control (survey + counting) with verified sunspot counts in progress.

C. H. Adams Data (1819–1823)

FARSuN has recovered 338 full-disk drawings and 1,056 daily entries (308 spot days, 748 spotless) from C. H. Adams (1819–1823), providing dense coverage with explicit spotless days. All spotless-day counts are cross-checked against Wolf's compilations, anchoring early-1800s calibration experiments.

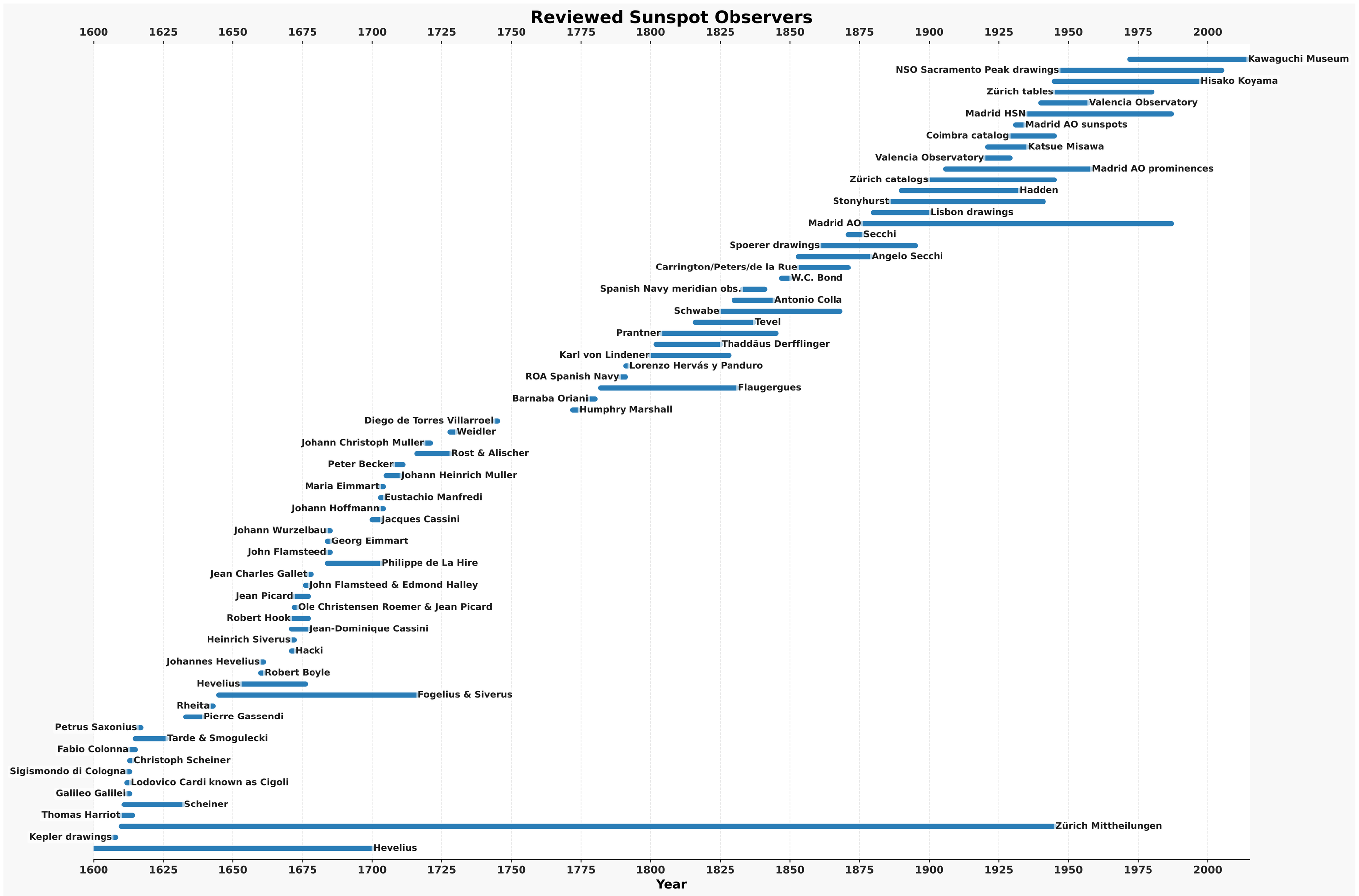
- Quality checks to align direct and reflection methods for consistent calibration.
- Orientation, and standardisation of drawings to enable positions and areas extraction.

Key Datasets

Augustin Stark Data (1813–1835)

The Stark dataset (1813–1835), comprising printed German descriptions of sunspot groups and chains, has been partially retrieved. Preliminary OCR and extraction tests show good potential, and ongoing work focuses on refining consistency, quality control flags, and metadata integration into the FARSuN database. Trial extraction campaigns continue so that descriptive passages are tagged consistently before bulk ingestion.

Community Data Recovery Efforts Since 2010



A non-exhaustive set of early telescopic sunspot datasets recovered and published since 2010, showing their temporal coverage from the early 1600s to the modern era. These community efforts—including major contributions by Hayakawa, Arlt, Vaquero, Carrasco, and others—have greatly expanded the number of available raw observations, enabling independent cross-checks of overlapping observers.

From Raw Sunspot Data to S_N V3.0

1. Focus on digitizing the most difficult periods, where it most meaningfully improves the final series.
2. Curate daily counts with provenance-rich metadata (observer, instrument, location, context).
3. Model how each observer counted sunspots, and use this to adjust their data so observations from different centuries can be compared on the same scale.
4. Aggregate to monthly, yearly, and hemispheric indices when positions are available with propagated uncertainties.
5. Benchmark against S_N V2.0 and long-term proxies (e.g. sunspot areas, magnetic flux, F10.7) to validate drift corrections.

Space Climate Impact

Citizen science
Interactive portal to mobilise volunteers to extract complex data from images and manuscripts.

Forecasting
Bias-free S_N V3.0 will improve solar cycle prediction skills and operational space weather baselines.

Climate links
Validated historical forcing refines long-term climate models and Earth radiation budget studies.

Open science
FAIR-compliant releases make the data easy to find and use, with online access tools and clear uncertainty estimates.

Poster QR Code

For more information on FARSuN project, datasets, and a digital copy of the poster, scan the QR code.

Contact: silso.info@oma.be

Data Portal: https://www.sidc.be/silso/